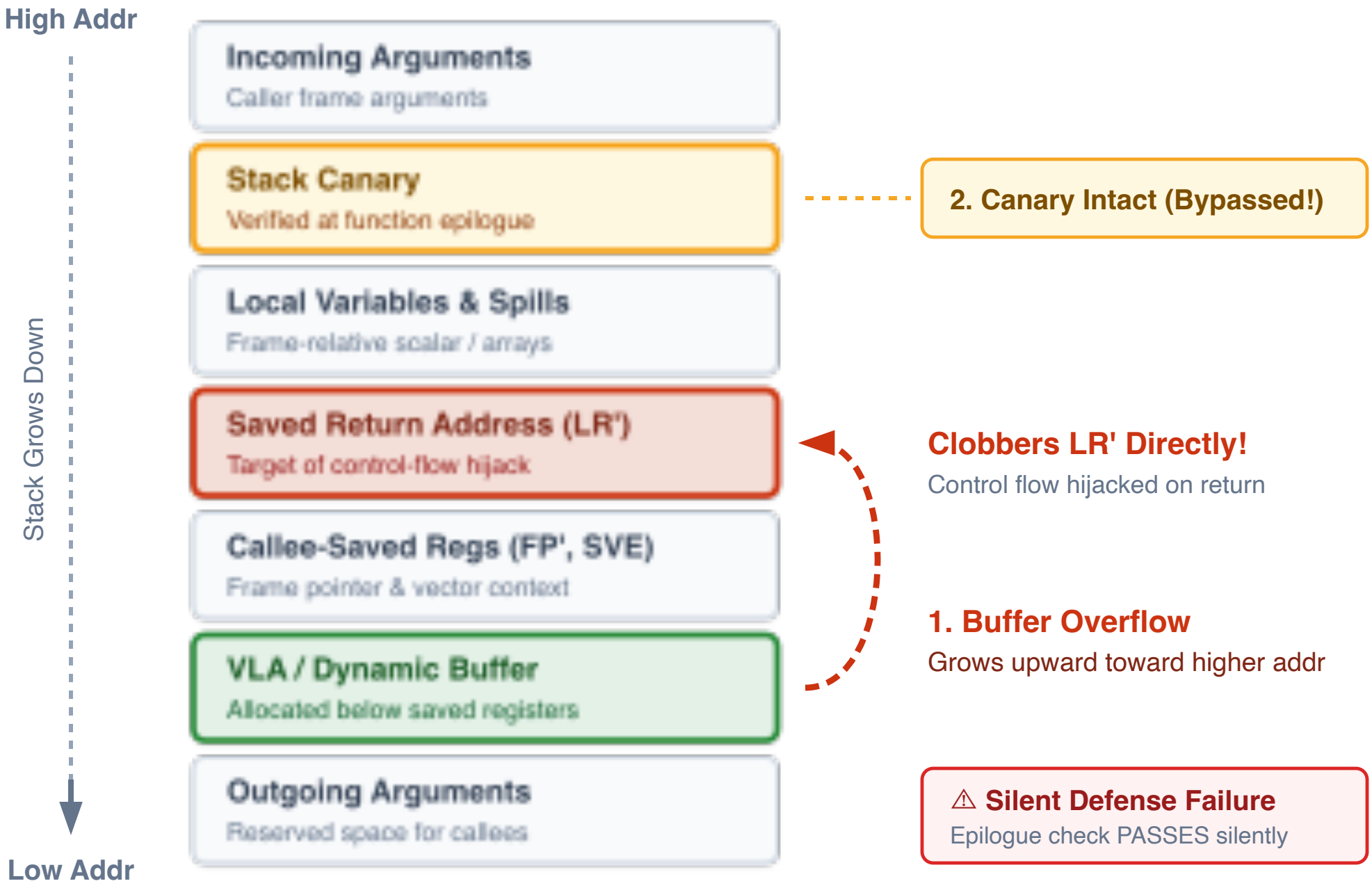

High Addr
Stack Grows Down
Low Addr
Incoming Arguments
Caller frame arguments
Stack Canary
Verified at function epilogue
Local Variables & Spills
Frame-relative scalar / arrays
Saved Return Address (LR')
Target of control-flow hijack
Callee-Saved Regs (FP', SVE)
Frame pointer & vector context
VLA / Dynamic Buffer
Allocated below saved registers
Outgoing Arguments
Reserved space for callees
2. Canary Intact (Bypassed!)
Clobbers LR' Directly!
Control flow hijacked on return
1. Buffer Overflow
Grows upward toward higher addr
⚠ Silent Defense Failure
Epilogue check PASSES silently